

Regular Article

Yokukansan Ameliorates Hippocampus-Dependent Learning Impairment in Senescence-Accelerated Mouse

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Yokukansan (YKS) is a traditional Japanese herbal medicine. It has been currently applied for treating behavioral and psychological symptoms of dementia in Japan. We investigated the effect of YKS on learning ability, hippocampal cell proliferation, and neural ultrastructural features in senescence-accelerated mouse prone 8 (SAMP8), a proposed animal model of Alzheimer's disease. Five-month-old male SAMP8 mice were randomly assigned to control and experimental groups. The control group had drug-free water *ad libitum*. The experimental mice were given 0.15% aqueous solution of YKS orally for eight weeks. Learning ability was assessed in Morris water maze test. Hippocampal cell proliferation was investigated using bromodeoxyuridine immunohistochemical method. The neural ultrastructural features, including myelin sheath and synapse, were investigated electron microscopy. Administration with YKS improved the hippocampal cell proliferation in dentate gyrus, and ameliorated learning impairment in SAMP8 mice. Numerous lipofuscin inclusions were presented in hippocampal neurons of the control mice. However, little were found after treatment with YKS. Myelin sheath was thicker and postsynaptic density length was longer after treatment with YKS. Administration with YKS ameliorated learning impairment in SAMP8 mice, mediated at least partially *via* delaying neuronal aging process, neurogenesis, myelin sheath and synaptic plasticity in the hippocampus. These results suggest that YKS might be effective for preventing hippocampus-dependent cognitive deficits with age.

Key words yokukansan; hippocampus; learning ability; myelin sheath; neurogenesis; synapse

Dementia is one of the most common neurocognitive diseases with age, characterized by memory loss, cognitive dysfunction, and decreased QOL. It is estimated that 47 million people live with dementia in the world.¹⁾ With the growth of aging population, dementia has become a global health and socioeconomic importance. Although the pathogenesis of dementia is still not well understood, researches have indicated that the hippocampus is a key target for preventing and treating this disease.^{2,3)} The hippocampus plays a pivotal role in processes of spatial learning and memory, which is one of the current issues for elucidating mechanisms of dementia.⁴⁾

The clinical effect of traditional medicine has been reevaluated with applied for treating diseases, including dementia.⁵⁾ Yokukansan (YKS) is a traditional Japanese herbal medicine, which has been approved by the Japanese government for the treatment of neurosis, insomnia, and children's night crying and irritability. It could also improve the behavioral and psychological symptoms of dementia.^{6–10)} It has been indicated that YKS is effective and well tolerated in dementia patients with behavioral and psychological symptoms without severe adverse effects. Recent studies showed that YKS has neuroprotective effects, promotes neuroplasticity,¹¹⁾ and ameliorates learning impairment.^{12–17)} However, the underlying mechanism of YKS on the amelioration of behavioral and psychological symptoms is not yet completely elucidated.

It was reported that senescence-accelerated mouse prone-8 (SAMP8) mice revealed age-related behavioral alterations from 4 months of age.¹⁸⁾ Hippocampal-dependent learning

ability began to decline in these mice as early as 2 to 4 months, as shown by impaired performance in the water maze test.¹⁹⁾ SAMP8 mouse has been used as a model of Alzheimer's disease, the most common type of dementia.^{20,21)} The present study investigated the effect of YKS administration on hippocampal neurogenesis, neuronal ultrastructural features, myelin sheath, synapse, and learning ability in SAMP8 mice.

MATERIALS AND METHODS

YKS YKS contains a mixture of seven medical herbs, which are registered in the Pharmacopeia of Japan ver. 17. The dried extract powder of YKS was supplied by Tsumura & Co. (Tokyo, Japan). Table 1 shows the herbal constituents and contents of YKS. The dried extract power of YKS was

Table 1. Herbal Constituents and Contents of Yokukansan

Botanical plant name	Contents (g)
Attractylodes lancea rhizome	<i>Attractylodes lancea</i> DE CANDOLLE 4.0
Poria sclerotium	<i>Poria cocos</i> WOLF 4.0
Cnidium rhizome	<i>Cnidium officinale</i> MAKINO 3.0
Angelica radix	<i>Angelica acutiloba</i> KITAGAWA 3.0
Uncaria uncis cum ramulus	<i>Uncaria rhynchophylla</i> MIQUEL 3.0
Bupleurum radix	<i>Bupleurum falcatum</i> LINNÉ 2.0
Glycyrrhizae radix	<i>Glycyrrhiza uralensis</i> FISHER 1.5

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produced in accordance with the formulation reported previously.^{7,10)} The components of YKS has been identified by three-dimensional HPLC (see Supplementary Fig. 1). The representative compounds include ferulic acid, glycyrrhizin and saikosaponin b2.¹⁰⁾ The extract quality is standardized on the basis of Good Manufacturing Practices defined by the Japanese Ministry of Health, Labor and Welfare.¹⁰⁾

Animals and Experimental Design Male SAMP8 mice at 5 months of age were obtained from Japan SLC, Inc. (Shizuoka, Japan). Five mice per cage were housed under controlled temperature ($23\pm1^{\circ}\text{C}$), humidity ($55\pm2\%$), and light (12h light/dark cycle, lights on at 06:00 and off at 18:00). We used 36 mice in the present study. The mice were maintained on a standard rodent chow (CE-2, CLEA Japan, Inc., Tokyo, Japan) *ad libitum*. This study was approved by the ethics committee of Asahi University School of Dentistry. All experiments were in compliance with the guidelines for laboratory animal care and use of Asahi University. The mice were randomly assigned to control and experimental groups. In the experimental group, YKS of 0.15% aqueous solution was treated orally for eight weeks. The concentration of YKS was determined by body weight and the average daily water intake of each mice. The dosage of YKS was decided based on the previous studies.^{13–15)} The control group had drug-free water *ad libitum*.

Morris Water Maze Test The water maze was carried out for both control and experimental mice ($n=6/\text{group}$) as reported previously.^{22,23)} A circular stainless pool (90cm in diameter and 30cm high) was filled with water (*ca.* 23°C) to a height of 23cm. A platform (12 $\times$ 12cm) was submerged 1cm under the water surface in the center of the tank. Mice were placed into the water from one of four randomly selected positions around the pool, and given 4 acquisition trials per day for 5d continuously. Escape latency and swimming path were recorded and analyzed with the aid of a software (Moveer/2D, Library Co., Ltd., Tokyo, Japan). All animals underwent a visible probe test 2h after the last training trial on the last day of training.

Bromodeoxyuridine (BrdU) Treatment We examined the hippocampal newborn cell proliferation in the dentate gyrus (DG) region after intraperitoneal injection of BrdU (50mg/kg; 10mg/mL dissolved in 0.9% sodium chloride, Sigma-Aldrich, St. Louis, MO, U.S.A.) into the mice ($n=7/\text{group}$) 5 times at 3-h intervals.²²⁾ The next day after the last injection of BrdU, mice were perfused *via* the ascending aorta with 0.9% sodium chloride followed by 4% paraformaldehyde solution under anesthesia (sodium pentobarbital 40mg/kg, intraperitoneally (i.p.)). The brains were carefully dissected from the skull and immersion fixed in 2% paraformaldehyde solution for 24h at 4°C .

BrdU Immunohistochemistry The hippocampal coronary sections (40 μm thick) were cut on a cryostat (CM1850, Leica, Wetzlar, Germany). Immunohistochemical detection was performed by using the avidin–biotin complex method.²⁴⁾ The brain sections were immersed in phosphate buffered saline (PBS), treated with 1% H_2O_2 for 10min, and then incubated with 5% normal goat serum for 1h at room temperature. Slices were incubated in primary antibody rabbit anti-BrdU (1:200, Abcam, Cambridge, U.K.) for 48h at 4°C . They were then incubated in secondary antibody biotinylated goat anti-rabbit immunoglobulin G (IgG) (Dako Cytomation, Glostrup,

Denmark) for 2h at room temperature. Slices were treated with peroxidase-conjugated streptavidin (Dako Cytomation) for 1h. The bound complex was visualized using 3,3'-diaminobenzidine. We used normal rabbit IgG instead of the primary antibody for a negative control.

Quantification of BrdU-Positive Cells To quantify the density of BrdU-positive cells in the hippocampal DG region, every fourth section of the series was chosen, and 8 sections (Bregma -2.12 to -6.30mm) per mouse were used for quantification analyses under a light microscope (Olympus BX-50, Olympus Corporation, Tokyo, Japan), with the aid of an unbiased stereological estimation.²⁵⁾ The numbers of BrdU-positive cells in the hippocampal DG region were counted by an investigator blinded to treatment group assignment, utilizing an image analysis software (Lumina Vision, Mitani Co., Ltd., Fukui, Japan). More than 50 BrdU-positive cells were counted in each brain.

Transmission Electron Microscopy After deep anesthesia, mice ($n=5/\text{group}$) were perfused *via* the ascending aorta with 0.9% physiological saline followed by Karnovsky's fixative (2% paraformaldehyde and 2.5% glutaraldehyde in 0.1M phosphate buffer, pH 7.4), as reported previously.²³⁾ The brain was carefully dissected and further fixed in the same fixative solution for 24h at 4°C . The specimens were postfixed in 1% OsO_4 for 1h. After dehydrating through an ascending graded acetone series, specimens were embedded in epoxy resin. The ultrathin sections (80nm thick) were obtained with glass knives on a Porter Blum MT-1 ultramicrotome (Ivan Sorvall, Inc., Norwalk, CT, U.S.A.) and collected onto copper mesh grids. The sections were observed using a transmission electron microscope (JEM-1400Plus, JEOL Ltd., Tokyo, Japan), after staining with 0.1% uranyl acetate and lead salts.

Myelin sheathes in the hippocampal Cornu Ammonis (CA) 1 region were observed at 10000 $\times$ magnification. Twenty images containing 200 axons per mouse were obtained for quantitative evaluation. The G-ratio (the ratio of the inner to the outer diameter of the myelin sheath) was determined, as previously described.^{23,26)} Synaptic structural analyses were performed at 30000 $\times$ magnification. Synapses were confirmed by clearly visible synaptic vesicles and postsynaptic density (PSD). Fifty synapses per mouse were selected for measuring PSD length, as previously described.^{23,27)}

Statistical Analysis All values are expressed as means $\pm$ standard deviation (S.D.) The statistical analysis was performed by using SPSS version 22. Statistical significance was determined using a Wilcoxon signed-rank test or Mann–Whitney test. *p* value of less than 0.05 was considered to be statistically significant.

RESULTS

Morris Water Maze Test All mice demonstrated improved performance during acquisition based on the decrease in the escape latency and swimming path length over the five training days (Fig. 1). On day 1, there was no significant difference between the control and experimental groups regarding the latency and path length, indicating both groups have similar motor and visual capabilities. Compared with the control group, the escape latency and the swimming path length were significantly shorter in the experimental group for days 3 to 5. We did not find any significant differences between

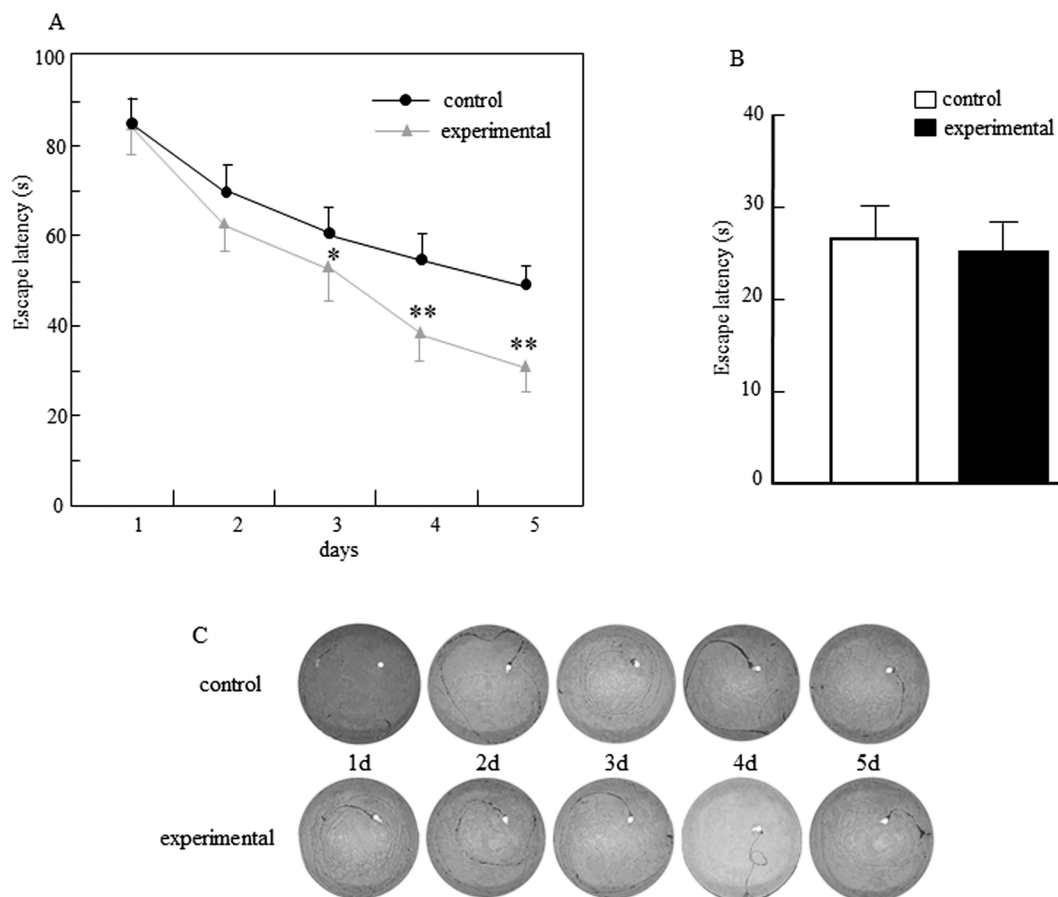


Fig. 1. Spatial Learning Ability in the Morris Water Maze Test

The escape latency in the experimental group was significantly shorter than that in the control group for days 3 to 5 (A). There was no significant difference between the control and experimental groups regarding the visible probe test, indicating both groups had similar motor and visual capabilities (B). The swimming path length of the experimental group was shorter than that in the control group (C). * $p < 0.05$. ** $p < 0.01$.

the control and experimental groups with regard to the performance in the visible probe test, indicating both groups had similar motor and visual capabilities (Fig. 1).

Hippocampal Cell Proliferation in the DG Region Representative photomicrographs of BrdU immunohistochemistry in the hippocampus DG region are shown in Fig. 2A. BrdU-positive cells were observed within the subgranular zone of the DG region in both control and experimental groups. The number of BrdU-positive cells of the experimental group was around 40% higher than that of the control group ($p < 0.01$, Fig. 2B). This finding suggests that treatment with YKS improved the hippocampal newborn cell proliferation in SAMP8 mice.

Ultrastructural Feature of the Hippocampal CA1 Neurons The ultrastructural features of the hippocampal CA1 neurons were examined using transmission electron microscope. The neuron of both control and experimental groups exhibited conspicuous nucleus with prominent euchromatin (Fig. 3). The cytoplasm contained cisternae of the rough endoplasmic reticulum, Golgi apparatus, and mitochondria with intact cristae. There were a lots of lipofuscin inclusions in the control group (Fig. 3). Lipofuscin inclusions were found in a great many varieties, small or large, round, oval or irregular. However, little were observed in the experimental group (Fig. 3). We did not find any obvious ultrastructural alterations of other organelles for both groups.

Myelin Sheath The ultrastructural features of the my-

elin sheaths were examined under the electron microscope. The morphology of axons in the control and experimental groups appeared grossly normal (Fig. 4A). The control group showed various irregular profiles of the myelin sheaths. Some were loosely organized with a disordered texture, and others exhibited stratification, collapse, or disruption, as well as disordered arrangements (Fig. 4A). Myelin sheaths in the experimental group appeared normal, exhibiting concentric compact multilamellar structure around the axon (Fig. 4A). The thickness of the myelin sheaths correlated with the axon diameter. Morphometric analysis showed that the G-ratio differed significantly between groups. Compared with the control group, the G-ratio was significantly lower in the experimental group, indicating thicker myelin sheaths in the experimental group (Fig. 4B).

PSD Length Figure 5 shows the representative electron micrographs of the hippocampal synapses in CA1 region. Compared with the control group, the estimated PSD length was significantly longer in the experimental group. There were no significant differences between the groups regarding the synaptic vesicles and synaptic cleft between the control and experimental groups.

DISCUSSION

In this study, we provide the first evidence that treatment with YKS for 8 weeks ameliorates hippocampal morpho-

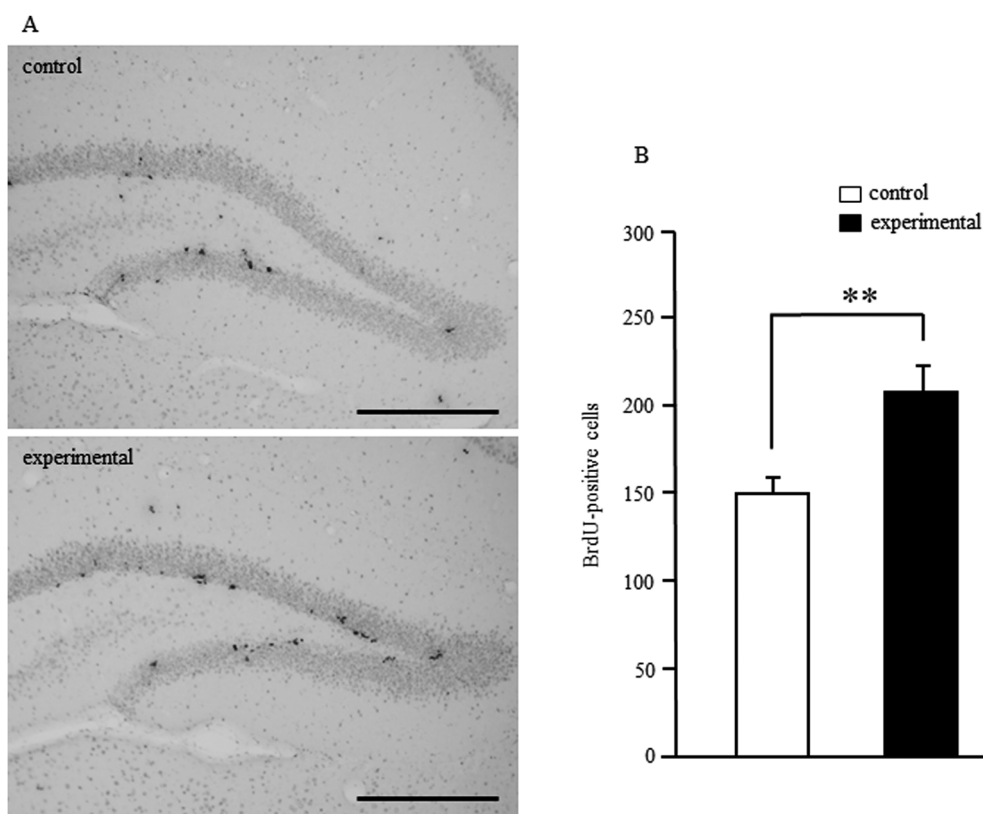


Fig. 2. The Hippocampal Newborn Cell Proliferation in the Dentate Gyrus

Light micrographs showing BrdU-positive cells (A) and the number of BrdU-positive cells in the control and experimental groups (B). The number of BrdU-positive cells in the experimental group was significantly higher than that of the control group. Scale bars, 100 μ m. ** $p < 0.01$.

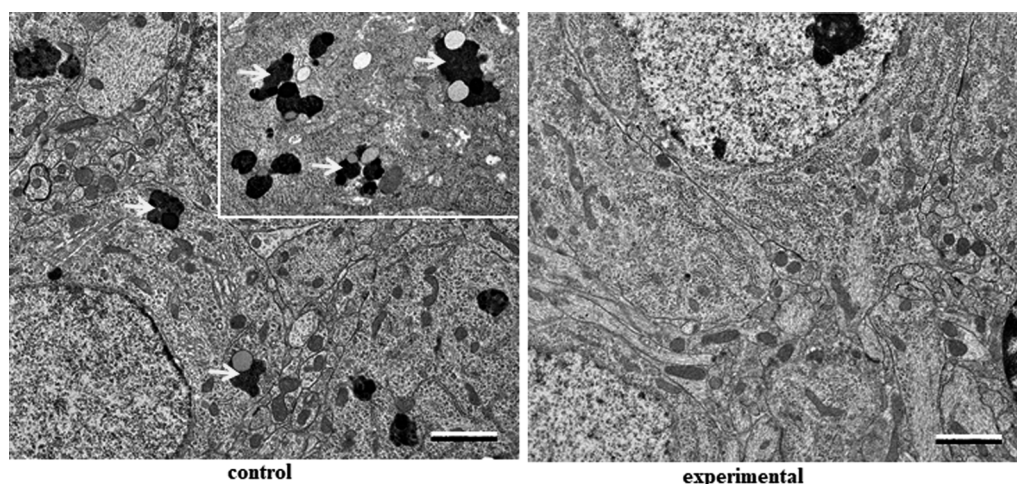


Fig. 3. The Neuronal Ultrastructural Features in the Hippocampal CA1 Region

There were numerous lipofuscin inclusions (arrows) in the cytoplasm of the control group. However, little lipofuscins were found in the experimental group. Scale bars, 2 μ m.

logical changes and learning impairment in SAMP8 mice. SAMP8 mice exhibit age-related learning impairment and rapid advancement of senescence.^{18,21,28} We found that treatment with YKS ameliorated learning impairment in SAMP8 mice. YKS was reported to prevent cognitive disturbances in Tg2576 mouse, the transgenic model of Alzheimer's disease,^{13,14} intracerebroventricular amyloid β protein ($A\beta$)-injection mice,^{16,29} animal models of schizophrenia,^{30,31} and aged rats.³²

Previous animal studies suggested that YKS could amelio-

rate hippocampal neurogenesis associated with cognition.³³ Our study showed that the hippocampal neurogenesis in SAMP8 mice was maintained by administration with YKS. Several studies confirmed the correlation of the newborn neurons in the hippocampal DG region with the spatial learning ability.^{34,35} Hippocampal neurogenesis is related to the formation of associative memories. Animals with impaired hippocampal neurogenesis perform worse than the control in hippocampal-dependent forms of spatial learning test.^{36,37} Therefore, the relationship between hippocampal neurogenesis

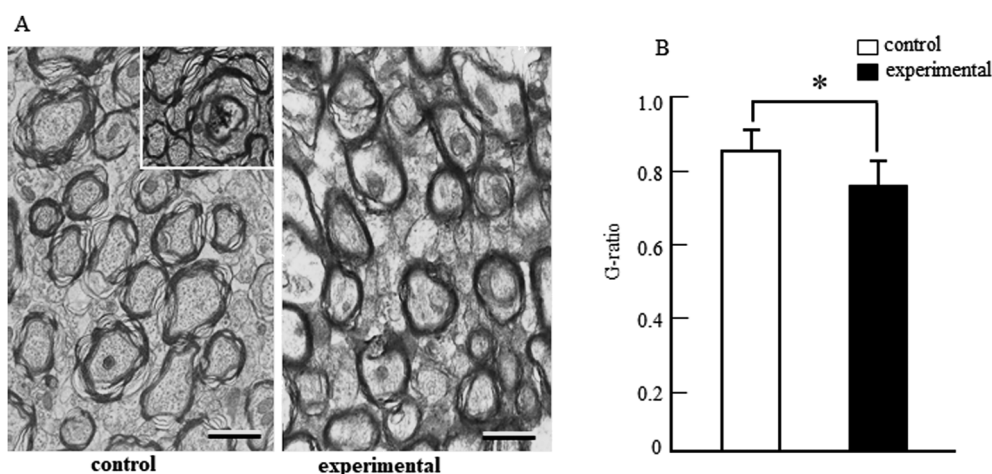


Fig. 4. Representative Electron Micrographs of the Myelin Sheaths (A) and G-Ratio (B) in the Hippocampal CA1 Region

The control group showed irregular profiles of the myelin sheaths. Some were loosely organized with a disordered texture, and others exhibited stratification, collapse, or disruption, as well as disordered arrangements. Myelin sheaths in the experimental group appeared normal, exhibiting concentric compact multilamellar structures around the axons. The G-ratio in the experimental group was lower than that of the control group. Scale bars, 1 μ m. * p <0.05.

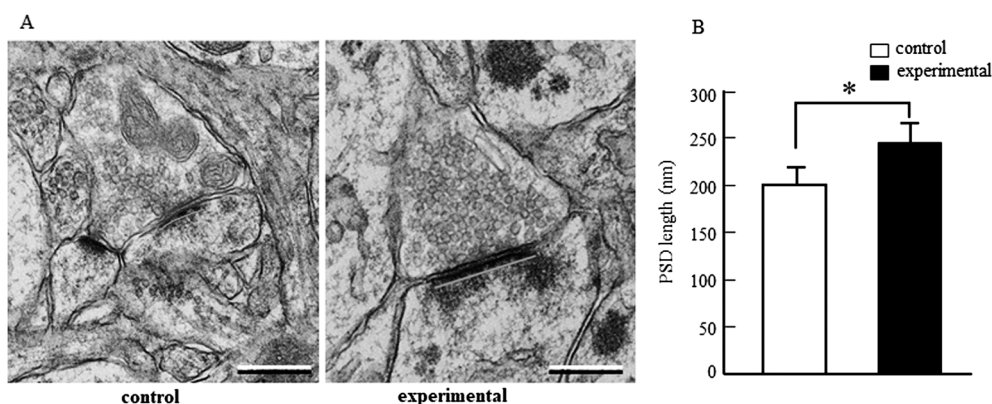


Fig. 5. Transmission Electron Micrographs of the Hippocampal Synapses (A) and Postsynaptic Density Length (B)

Postsynaptic density lengths are indicated with lines. Postsynaptic density length in the experimental group was longer than that of the control group. Scale bars, 100 nm. * p <0.05.

and spatial learning ability indicates that preserved hippocampal neuronal proliferation after treatment with YKS is involved in the maintenance of learning ability in SAMP8 mice. It was reported that the number of BrdU-positive cells reduced in the hippocampal DG region of the aged rats, and this decrease was improved by YKS treatment, which may influence the proliferation of neural stem cells in the hippocampal DG region.³³⁾ It is conceivable that YKS has pharmacological potency for preserving the hippocampal neurogenesis in aged animals.

Lipofuscin accumulations of the hippocampal neurons are commonly found in aged animals.^{38,39)} Lipofuscin accumulations were observed in the hippocampal neurons of SAMP8 mice, especially after treatment with kainic acid. It was considered that SAMP8 mice were susceptible to kainic acid-induced oxidative damage, which might be one of the causal factors in lipofuscin accumulation of the hippocampal neurons.³⁸⁾ The present study confirmed that numerous lipofuscin inclusions were presented in the hippocampal neurons of the control SAMP8 mice. However, there were little lipofuscin inclusions after administration of YKS. We consider that YKS might play a role in delaying neuronal aging process, involved in lipofuscin accumulations. However, direct evidences for the

contribution of lipofuscin accumulation, lipid peroxidation and protein oxidation to aging process are still unidentified. The precise mechanism of this phenomenon needs further investigation.

The myelin sheaths facilitate the rapid conduction of nerve impulses in myelinated fibers. Thinning or deformation of the myelin sheaths leads to a decrease in nerve conduction velocity.²⁶⁾ The hippocampal abnormal myelin sheaths are involved in behavioral impairments, inducing delayed cognitive development.⁴⁰⁾ Accordingly, we consider that prevention of learning impairment in SAMP8 mice by treatment with YKS may be associated with the maintenance of the hippocampal myelin sheaths. Previous study showed that oligodendrocytes in the hippocampal CA1 region decreased progressively with age.⁴¹⁾ Myelin sheaths of the central nervous system are formed by oligodendrocytes. We recently found that both myelin sheaths and oligodendrocytes in the hippocampus were influenced by prenatal stress and chewing stimulation.²³⁾ We presume that the alteration of myelin sheath is associated with age-dependent degeneration of oligodendrocytes in SAMP8 mice, though the oligodendrocytes were not assessed in the present study.

Synapse is a highly specialized structure designed to guar-

antee precise and efficient communication among neurons.⁴²⁾ The synaptic structural plasticity reflects the physiological function and plays a critical role in spatial learning ability.⁴³⁾ Synaptic morphological changes, including the PSD length is closely related to synaptic functional plasticity. Furthermore, synaptic functional alterations are always accompanied by the structural changes.⁴⁴⁾ PSD contains a lot of receptors, scaffolding proteins, and signaling complexes involved in synaptic plasticity and transmission. It was reported that induction of long-term potentiation is associated with an increase in PSD length.⁴⁵⁾ Synaptic size can be determined by the measurement of PSD length, which is an important factor related to the synaptic plasticity and neurobehavioral activities.⁴⁶⁾ Our study showed that administration with YKS increased the PSD length in the hippocampal CA1 region of SAMP8 mice. We speculate that the amelioration of learning and memory impairment by YKS involved in the alteration of the hippocampal PSD length in SAMP8 mice.

In conclusion, our results indicate that treatment with YKS for 8 weeks prevents age-related learning impairment in SAMP8 mice, mediated at least partially *via* delaying neuronal aging process, neurogenesis, myelin sheath, and synaptic plasticity in the hippocampus. Further studies are needed to clarify the detailed mechanisms underlying the neuroprotective effects of YKS.

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Conflict of Interest The authors declare no conflict of interest.

Supplementary Materials The online version of this article contains supplementary materials.

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